# A3.1 Diversity of organisms

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a3-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 118–124 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

## What counts as a species?

### A3.1.12 Limits of the species concept

pp. 118–120

The biological species concept depends on interbreeding. Asexual organisms produce clones and never interbreed, so each line could count as a separate species. Bacteria also pass genes sideways by **horizontal gene transfer**, even between distant groups, so their gene pools are not isolated.

### A3.1.13 Chromosome number

p. 120

#### Chromosome numbers and hybrids

A3.1.13



Human karyotype: 46 chromosomes. NHGRI, public domain.

| Species | Chromosomes (2n) |
|---|---|
| Human | 46 |
| Chimpanzee | 48 |
| Horse | 64 |
| Donkey | 62 |
| Mule (hybrid) | 63 |

**Try it**

Horse gametes have 32 and donkey gametes 31. Why can a mule grow by mitosis but not make gametes by meiosis?

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Members of a species share one chromosome number. A hybrid of parents with different numbers, such as a mule (63), has chromosomes that cannot all pair in meiosis, so it is usually infertile.

# Identifying species

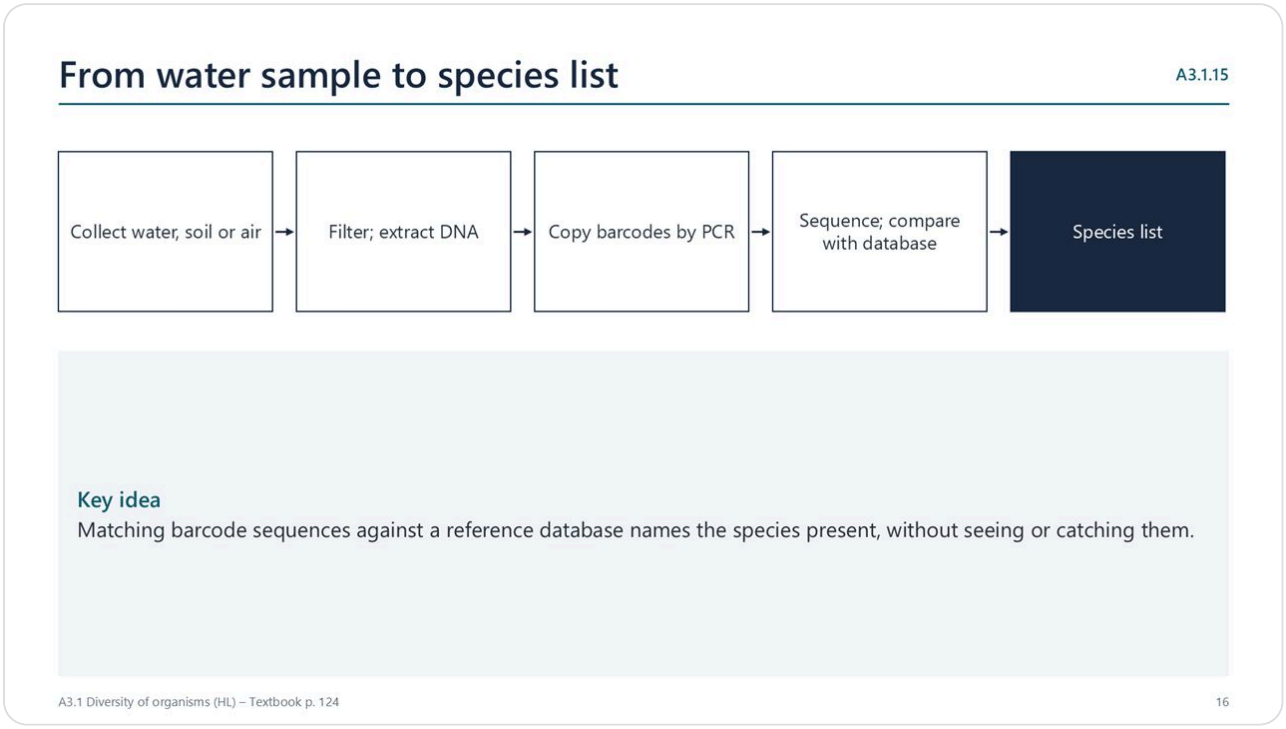
## A3.1.14 Dichotomous keys

pp. 121–123

A key is a series of **couplets**: pairs of opposite statements about one feature. Use features that are easy to see and do not vary, and check that every route ends at one species. Test your key by swapping it with another group.

## A3.1.15 eDNA and barcodes

p. 124




Organisms shed DNA into water, soil and air. Barcode regions, short standard sections that differ between species, are copied by PCR, sequenced and matched to a database. This shows quickly which species are present, but not how many individuals there are.

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